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Gendered Labor Market Responses to SNAP Work Requirements

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Abstract

Advocates of work requirements believe they encourage return-to-work rather than public dependency, while critics argue that they unfairly burden the most vulnerable. Since 1996, the federal food stamp program has used work requirements conditions for benefit eligibility, with individuals classified as Able-Bodied Adults Without Dependents (ABAWD) facing more stringent work requirements relative to the rest of the eligible population. If ABAWDs fail to meet the 20 hours a week work requirement, the benefits they receive are restricted to three months in three years. Due to high unemployment, states can receive work requirement waivers, and the entirety of the United States (US) received a waiver after the Great Recession (2008). Resulting from the post-recession recovery, ABAWD work requirements have been increasingly implemented across the US; by the end of 2016, 40 states had implemented work requirements. The paper studies whether ABAWD work requirements increase the labor supply of those impacted by their implementation, and whether these effects vary with income and gender. We use difference-in-differences and event studies to explore the impact of ABAWD work requirements on labor outcomes. We find evidence of nonlinearities in responses by income. The difference-in-differences results indicate that work requirements are associated with an increase in employment for persons earning between $7,500 and $15,000 annually, driven by women, and a possible decrease in labor force participation among the lowest income men. The event studies support these results but indicate that tradeoffs are likely occurring as positive effects are found among the $7,500-15,000 income group starting around the time of work requirement implementation followed by a lagged effect approximately six months after in the $15,000-$25,000 income bracket. These results suggest a gendered labor response to work requirements that varies with income, which could be used to improve targeting of work requirements towards those individuals whose labor market outcomes are more likely to respond positively.

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ACRONYMS

ABAWD	Able-Bodied Adult Without Dependents
ACA	Patient Protection and Affordable Care Act
ACS	American Community Survey
AFDC	Aid to Families with Dependent Children
BBCE	Broad-Based Categorical Eligibility
CPS	Current Population Survey
CPS – ASEC	Current Population Survey – Annual Social and Economic Supplement
DD	Difference-in-Differences
E&T	Employment and Training
FNS	Food and Nutrition Service
FPL	Federal Poverty Line
LSA	Labor Surplus Area
OLS	Ordinary Least Squares
PRWORA	Personal Responsibility and Work Opportunity Reconciliation Act of 1996
RDD	Regression Discontinuity Design
SNAP	Supplemental Nutrition Assistance Program
SSDI	Social Security Disability Insurance
SSI	Supplemental Security Income
TANF	Temporary Assistance for Needy Families
USDA	United States Department of Agriculture
USD	United States Dollars
US	United States

1. Introduction:

Although tying food assistance eligibility to employment has been standard policy in the United States (U.S.) for decades, such work requirements remain controversial. The United States food assistance program, the Supplemental Nutrition Assistance Program (SNAP), requires all recipients to make some effort towards re-entering the workforce, but limits receipt among adults between the ages of 18 and 49 without dependents (ABAWDs) to three months within three years unless they work at least 20 hours per week. At the federal level, support for food assistance-related work requirements varies by administration. Prior to the COVID-19 pandemic, the Trump administration had been considering augmenting SNAP work requirements, a policy that was predicted to disqualify 700,000 SNAP participants[2]. The Biden administration has since reversed the Trump administration's aforementioned SNAP proposals[3], but the topic continues to be a subject of public debate and the efficacy of work requirements in achieving their stated objective, increasing labor supply, remains an important policy matter. Despite the widespread use and controversy associated with SNAP work requirements, the academic literature only offers limited guidance. Work requirements have been shown to decrease SNAP participation (Harris, 2020; Gray et al., 2020; Ku et al, 2019, Ganong and Liebman, 2019) and have small positive effects on employment (Harris, 2020). This paper adds to that literature by studying the effects of work requirements on the decision to ever seek work, i.e., labor force participation, as well as analyzing the gendered effects of SNAP on employment across lower income subgroups. Using difference-and-differences and event studies approaches, we find positive effects on labor force participation and employment among individuals earning between $7,500 and $15,000 per year, driven by women, but these positive policy responses are offset by negative effects among those earning between $15,000 and $25,000, again particularly among women, with negative effects also found on labor force participation among men earning between $0 and $7,500.

SNAP, formerly and still colloquially known as "food stamps," provides food assistance to low or no-income individuals and households with the stated purpose of alleviating food insecurity for the poor and needy. In terms of the number of recipients, it is the largest public assistance program in the United States. SNAP participation increased rapidly as a result of the Great Recession and peaked in 2013, after which SNAP participation consistently declined until 2019. To give a notion of the scale of the program, for the fiscal year 2019, SNAP benefits worth 56 billion USD were distributed to about 36 million individuals (11 percent of the US population) or 18 million households[4]. Beginning 2020, SNAP

[2] https://www.foodandwine.com/news/snap-food-stamp-work-rules-trump-administration. Accessed 06/17/2021
[3] https://www.foodandwine.com/news/snap-benefits-food-stamps-work-requirement-biden-administration. Accessed 06/17/2021
[4] https://fns-prod.azureedge.net/sites/default/files/resource-files/34SNAPmonthly-9.pdf. Accessed 10/05/2021

participation increased due to the economic turmoil and ensuing recession caused by the COVID-19 pandemic. During this period, SNAP formed an important component of the social safety net. As a result, in the year 2021, SNAP benefits worth 80 billion USD were distributed to about 42 million individual enrollees (12.5 percent of the population) and 22 million households.[5] Currently, SNAP benefits are distributed to recipients via an electronic debit card that can be swiped at a store checkout similar to any other bank debit card and SNAP benefits apply for most grocery store food items.

The stated purpose of policymakers for associating work requirements with program participation (workfare) is to decrease welfare dependency among the poor and incentivize them to rejoin the work as soon as they are able. For some the policy likely has its intended effect and causes enrollees to increase their labor supply, but for others, the increase in transaction costs may force them off SNAP and onto non-workfare assistance, such as private, nonprofit food banks. For those for whom finding work is challenging or for whom the return from work is low may choose to opt out of the labor force altogether. This low-income group generally has tenuous links to the labor market and may only have a nominal presence in the labor force, if at all. Given that implementation of work requirements will lead to a discontinuation of SNAP benefits as a consequence of their inability to meet the more stringent work requirements, irrespective of their labor market presence, some may choose to exit the labor force altogether, perhaps eventually obtaining assistance through other public and private assistance programs, such as Supplemental Security Income (SSI), which does not have a work requirement.[6]

Recognizing that work requirements may be unduly burdensome in times of economic downturn, the implementation of work requirements is tied to local labor market conditions. The Great Recession-induced labor market effects triggered waivers of work requirements in nearly all states. As the economy recovered, work requirements were reinstated in states across the US, with the greatest number reinstated in 2016. By 2015, 21 states had work requirements; by the end of 2016, 40 states had work requirements. Unfortunately, when labor market conditions are good and unemployment is low, only the least employable are likely to remain unemployed and on public assistance.

The purpose of this study is to determine whether ABAWD work requirements positively affect the labor supply and employment of those impacted by their implementation and whether the effect varies by income and gender. We leverage the 2016 implementation of work requirements in difference-in-differences (DD) and difference-in-difference-in-differences (DDD) identification strategies to estimate the impact of work requirements on labor force participation and employment. As the poorest individuals, even among SNAP recipients, are likely to have the fewest resources and face the worse labor market

[5] https://fns-prod.azureedge.net/sites/default/files/resource-files/34SNAPmonthly-9.pdf. Accessed 10/05/2021

[6] Stith (2021) finds that SNAP-related work requirements are associated with an increase in Supplemental Security Income (SSI) receipt.

prospects, we test whether effects vary by income. Furthermore, as women may be more likely to participate in public assistance programs (Irving and Loveless, 2015), we predict that women will be more likely to meet the new requirements to maintain access to SNAP benefits, while men are more likely to become discouraged and opt out of the workforce altogether.

While there has been an abundance of studies on labor force responses to the work requirements associated with Temporary Assistance to Needy Families (TANF), formerly known as Aid to Families with Dependent Children (AFDC) (Moffitt, 1999; Ziliak et al., 2000; Grogger and Michalopoulos, 2003; Grogger, 2003; Grogger, 2004), research on SNAP work requirements and their impact on labor market outcomes has been more limited.

In the ABAWD work requirement and labor outcome literature, the majority of the literature finds a null effect of work requirements on employment. Most studies use data on employment from the Current Population Survey (CPS) and American Community Survey (ACS), and find very limited effects on employment overall. Cuffey et al. (2015), using information about ABAWD work requirement waivers from 2005-2009 and a triple-difference approach, finds no evidence that newly waived adults changed their work patterns in response to the program change. Another working paper by Stacy, Scherpf, and Jo (2018) found no employment effect among SNAP recipients. Han (2020) exploits area-time-variation in SNAP work requirement waivers and 15 percent exemptions to estimate the impact on employment and finds no noticeable effect, concluding that work requirements are unlikely to increase employment. In the study closest to ours, Harris (2020) studies work requirements implemented between 2011-2017 and finds small positive employment effects that do not vary by gender. A study using administrative earnings data from Virginia and a Regression Discontinuity Design (RDD) finds a null effect of work requirements on employment, but some suggestive evidence of an increase in earnings. They speculate that many SNAP participants have low or non-existent connection to the labor market and consequently are unlikely to demonstrate employment responses which would dilute the average employment effect (Gray et al., 2021).

Although work requirements cause some individuals to lose food assistance (Harris, 2020; Gray et al., 2020; Ku et al, 2019), the effects of gaining food assistance on labor supply may be informative to this study as well. On the extensive margin, Hagstrom (1996) found food assistance decreased employment among married couples and Hoynes and Schanzenbach (2012) found decreased employment among single parents. Fraker and Moffitt (1998) found no effect on the extensive margin (employment) but did document a nine percent reduction in hours worked. These studies address slightly different populations than our study, but if the results from losing food assistance mirror those from gaining food assistance, these studies suggest that work requirements, documented to decrease food assistance receipt, will lead to positive employment effects.

To summarize, the existing literature shows negative effects on employment from public assistance enrollment, including SNAP. Studies of work requirements associated with SNAP and other public assistance programs typically find positive or null effects on employment. Our study extends the literature by analyzing labor force participation; identifying nonlinearities in responses by income, which vary by gender; and using month-level rather than annual data. By focusing on the 2016 implementation, we are able to isolate the effect of more recent work requirements, which likely have greater policy relevance than those implemented in the more distant past. In addition, except for Gray et al (2021), all prior studies on the subject use a staggered difference-in-differences (DiD) identification without addressing concerns about the staggered treatment approach raised in the new econometric DiD Literature (Goodman-Bacon, 2019; Callaway and Sant'Anna, 2020; Sant'Anna and Zhao, 2020). We use a 2X2 difference-in-differences approach, avoiding issues with the staggered variant of the differences-in-differences approach.

The remainder of this paper is organized as follows. Section 2 presents the context and background of the research question. Section 3 presents a discussion on the data sources, study sample and descriptive statistics. In section 3, the empirical framework is presented, which involves a discussion of the empirical strategy, identification approach and sample construction. In section 5, the outcome trends, empirical results and robustness checks are presented. Section 6 concludes with policy recommendations, study limitations and suggestions for future research.

2. Background:

The Personal Responsibility and Work Opportunity Reconciliation Act of 1996 (PRWORA) reformed the welfare system in a manner that the creators hoped would incentivize re-entry of welfare recipients into the workforce and decrease what they perceived to be welfare dependency. Two important components of the welfare reforms were time limits and the use of work requirements to determine welfare eligibility. In the case of SNAP, a special work requirement component was added for the ABAWD category of individuals in addition to the pre-existing general work requirement. All SNAP recipients ages 16-59 are subject to the general work requirements, with some exceptions.[7] The general work requirements include registering for work, taking up employment if a suitable job has been offered and participating in SNAP Employment and Training (E&T) or workfare (if offered by the state) in case of unemployment. Other conditions include not voluntarily quitting a job or reducing work hours below 30 a week without a good reason.

The new work requirements established in the PRWORA introduced work requirement-associated time limits for ABAWD individuals. An ABAWD is a person between 18 and 49, not disabled

[7] https://www.fns.usda.gov/snap/work-requirements. Accessed 06/12/2021

(United States Department of Agriculture, Food and Nutrition Service) and with no dependents present in the household. An ABAWD is only able to receive food stamps for 3 months within a span of 3 years unless he/she meets certain special work requirements; this component is referred to the ABAWD time limit. The special work requirements entail working a minimum of 20 hours to be eligible for food stamps for more than 3 months in 3 years. If they meet the minimum work requirements and remain 130 percent below the Federal Poverty Level, ABAWDs will continue to receive SNAP benefits. Another way for individuals subject to the time limit to fulfill the associated ABAWD work requirement and maintain their SNAP eligibility is participation in a State SNAP Employment and Training (E&T). However, the provision of an Employment & Training program for recipients by states is optional, most have limited E&T programs, and states that do have robust E&T programs still do not cover everyone that is eligible.[8]

In acknowledgment of circumstances where the ABAWD requirements would be unduly burdensome due to insufficient job opportunities, the PRWORA allowed for ABAWD work requirements and the associated time limit to be suspended if certain conditions were met. According to the Food and Nutrition Service (FNS), the common criteria by which states can qualify for a time limit waiver is as follows: (1) A recent 12-month unemployment rate above 10 percent overall in the geographical unit; (2) A recent 3-month unemployment rate above 10 percent; (3) Designation as Labor Surplus Area[9] (LSA) by the Department of Labor; (4) Qualification for extended unemployment benefits; or (5) A recent 24-month average unemployment rate 20 percent above the national average for the same 24-month period. The time limit waiver entails ABAWD work requirements and the associated three-month time limit on SNAP benefits is not applicable in the region granted the waiver. The time limit waiver is the same irrespective of the condition or conditions met. The assessment for waivers is made every quarter, but when granted, the waivers are for the length of a year. At the end of the year the waiver is reassessed, and if the conditions for a waiver are not met, ABAWD work requirements are implemented. Once the ABAWD work requirements are reintroduced, an ABAWD is able to receive SNAP benefits for only 3 months within a span of 3 years if he/she does not work the required hours. The three-month count begins following the implementation of ABAWD work requirements.

States can apply for a state-wide or full ABAWD time limit waiver or partial state waiver for certain counties or geographical units within the state. Even if the state does not qualify for a waiver as a single geographical unit, it can apply for a geographical region within the state that does meet the requirements – we refer to this as a partial state waiver. During our sample period states have

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[8] https://www.clasp.org/sites/default/files/public/resources-and-publications/publication-1/2016-01-21-ABAWD-Work-Activities-Revised-FINAL.pdf. Accessed 06/22/2021

[9] A jurisdiction is classified as a labor surplus area when its average unemployment rate is at least 20 percent above the average unemployment rate for the nation during the previous two calendar years, with a ceiling of 10 percent and a floor of 6 percent for periods of very high or low national unemployment.

considerable flexibility in how they group counties or other geographical units together to create a single entity for an ABAWD waiver application.[10]

As a consequence of the Great Recession (2008), SNAP enrollment increased dramatically. Motivated by the dire economic conditions of the time, the 2009 Recovery Act suspended the ABAWD work requirements nationwide from April 2009 to September 2010. Despite the suspension being lifted in September 2009, all states except for one still qualified for ABAWD time limit waivers. Due to the ensuing recovery from the Great Recession and the resulting improvement in economic indicators, state work requirement waivers began expiring, and by the end of 2016, the ABAWD associated work requirement were in effect in 40 states. Of these 40 states, 21 states had ABAWD work requirements implemented for the first time in the second quarter of 2016. Of the 21 states, 5 fully implemented ABAWD work requirements, the largest and most recent single implementation of work requirements since the Great Recession, thus providing us with a labor market response for recently implemented work requirements, whose labor response may be more useful for predicting current policy effects than those implemented earlier. The five states that fully implementated ABAWD work requirements beginning the second quarter of 2016 coupled with non-continuous waiver criteria allow for a quasi-experimental design.

3. Data:

Our outcome variables and individual level explanatory variables come from the Current Population Survey (CPS) Basic Monthly Supplement; a monthly survey of U.S. households conducted by the U.S. Census Bureau for the Bureau of Labor Statistics (BLS). The sample space for our study encompasses the year preceding the second quarter of 2016 (April 1st, 2016) and the year after. The sample period covers 24 months from April 2015 through March 2017. We use the basic monthly component of the CPS, which includes labor force items for those aged 15 and above for the sample period. The basic monthly components are available for every month of the year. Full demographic information is collected for all household members. We use sampling weights from the basic monthly component of the CPS to make our sample representative of the population.

The Able-Bodied Adults Without Dependents (ABAWD) category defines an individual aged 18-49, who is not disabled, with no dependents present in household (USDA, Food and Nutrition Service). To match this definition as closely as possible, the sample space for our analysis is restricted to individuals of ages 18-49 with no biological child present in household. SNAP assistance is only available up to 130 percent of the federal poverty line, e.g., the SNAP income eligibility limit for October 2017 through September 2018 was $1,307 for a one-person household and $1,760 was $1,307 and for a

[10] We do not have county-level waiver information and consequently exclude states with partial waivers. Furthermore, respondent county identifiers are provided for only a subset of the CPS sample.

two-person household. We further restrict our sample to single-person households in order to avoid complications arising from household-level reallocation of labor which would require the use of alternate models to identify effects. We run our analysis for four income groups: households earning between $0-$7,500 (deep poverty), $7,500-$15,000 (middle poverty), $15,000-$25,000 (around the FPL) and the full sample encompassing $0-$25000 in order to break out how the effects may vary with degree of poverty. For example, for individuals in deep poverty, support may be more critical and labor market connections more tenuous thane for those closer to non-poverty. Because eligible women may be more likely to participate in public assistance (Irving and Loveless, 2015), we run additional sub-analyses by gender.

Information on work requirement waivers was obtained from United States Department of Agriculture, Food and Nutrition website.[11] Table 1A in the Appendix lists the 9 states without waivers for the entire sample period, the 7 states with waivers for the entire period, and the 5 states that received waivers in the second quarter of 2016. Table 1 provides descriptive statistics for the CPS variables by type of waiver regime for both the outcome variables and explanatory variables used in our specifications.

4. Empirical Framework:

We use the full implementation of ABAWD work requirement in the aforementioned five states as a quasi-experiment. The 5 states that experience full work requirements in the second quarter of 2016 serving as the switcher (treated) group, while the six untreated states with no ABAWD work requirements and the 7 states with full ABAWD work requirements for the entire period serve as two control groups. We drop the states with partial work requirement implementation from our analysis because our data source, the Current Population Survey (CPS) is missing county identifiers for nearly half of US counties, thus preventing us from matching county waivers to nearly half of the respondents in our data.

We partition our sample by before and after the second quarter of 2016 and use a difference-in-differences approach to capture the impact of the imposition of work requirements on labor supply.

4.1. Estimation Strategy: The paper utilizes variation in state SNAP-related work requirements to identify quasi-causal changes in single-person household labor supply decisions. Following is the functional form of the Ordinary Least Square (OLS) estimator utilized for regression analysis.

$$\text{Labor Outcome}_{\text{ism}} = \alpha + \gamma(\text{ABAWD}_{\text{sq}} * \text{Post}) + \delta(\text{ATABAWD}_{\text{sq}} * \text{Post}) + \text{ABAWD}_{\text{sq}} + \text{ATABAWD}_{\text{sq}} + \text{Post} + \text{X}_i + \pi_s + \lambda_s + \eta_m + \varepsilon \quad (1)$$

[11] https://www.fns.usda.gov/snap/ABAWD/waivers. Accessed 01/15/2018.

$\text{Labor Outcome}_{\text{ism}}$ represents our three (extensive margin) outcome variables: probability of participating in the labor Force {0,1} and Employment {0,1} for individual i in state s and month m.

ABAWD_{sq} is a dichotomous dummy variable indicating the state's status in regard to ABAWD-associated work requirements in state s and quarter q. The categories include "never treated" states in which work requirements were not implemented during the sample period and "switcher" states. $\text{ABAWD}_{\text{sq}} = 0$ if the states have no work requirements implemented during our sample period (never treated states) and $\text{ABAWD}_{\text{sq}} = 1$ if the state had work requirements implemented in the second quarter of 2016 (switcher states). The always treated states are captured by $\text{ATABAWD}_{\text{sq}}$. $\text{ATABAWD}_{\text{sq}} = 0$ if work requirements were not implemented during the sample period (never treated states) or if work requirements were implemented during the second quarter of 2016 (switcher states). $\text{ATABAWD}_{\text{sq}} = 1$ if the state had work requirements implemented for both the twelve months prior and beginning the second quarter of 2016 (always treated states). The post period variable is $\text{Post} = 0$ for the twelve months prior to the second quarter of 2016 and $\text{Post} = 1$ for the twelve months beginning the second quarter of 2016. The interaction between Post and $\text{ABAWD}_{\text{sq}} = 1$ (switcher states) provides the effect of ABAWD work requirements on the labor force outcomes of interest. The interaction between Post and $\text{ATABAWD}_{\text{sq}}$ (always treated states) serves as a falsification test; the always treated states function as a second control having experienced no intervention during the sample period, consequently there should be no demonstrable impact on the always treated. The policy variable (ABAWD_{sq}) and period variable (Post) are also added independently to the model.

We also include several control variables. The term X_i is a vector of covariates for individual or household level characteristics. We account for demographics (age, race/ethnicity and gender), education attainment, college attendance and central city location. π_s is a vector of state-level controls. The state-level controls include variables for TANF and Medicaid monthly caseloads, and similar to Harris (2020) and Ganong and Liebman (2018), we include a dichotomous variable for whether the state has implemented SNAP Broad-Based Categorical Eligibility[12] (BBCE) and a variable for the BBCE state-level gross income limits. In addition to allowing for categorical eligibility, BBCE policies allow states to raise the SNAP eligibility income threshold. We also include quarterly state GDP to account for economic conditions which affect both work requirement implementation and labor market outcomes. State fixed effects are given by λ_s and account for time-invariant unobserved determinants that vary by state. Month

[12] Broad-based categorical eligibility (BBCE) is a policy in which households may become categorically eligible for SNAP because they qualify for a non-cash Temporary Assistance for Needy Families (TANF) or state maintenance of effort (MOE) funded benefit.

fixed effects are given by η_m, which capture month-level factors, such as seasonality, which affect all states similarly. The state and month fixed effects are operationalized by a creating a dummy variable for each state and month. Standard errors are clustered at the state-level.

For the full sample period, the 24 months from April 2015 through March 2017, the CPS has 3,408,353 observations. When we restrict the sample to ABAWD individuals in single person households the sample observations decline to 206,597. When we impose a further restriction by limiting the sample to \$0-\$25,000 annual earners the observations are reduced to 65,627. Further dropping non-citizens because they are ineligible for welfare programs and states for which we do not have work requirement information (partial waiver states), the sample declines to 32,448. Lastly, 3,077 observations do not have sample weights provided, leading us to exclude these observations along with 41 respondents missing labor force participation information leaving us with a sample of 29,330, which is our regression sample size for the full income category in Table 2.

The interaction between $\mathrm{ABAWD_{sq}}$ (policy variable) and Post (period variable) gives the difference-in-differences estimator or coefficient of interest, γ, which represents the differential effect on switchers relative to the never treated states (the control group).

In order to evaluate possible pre-trends and policy leads and lags, we supplement our DD approach with event studies. Our approach is similar to the DD specification except that the {0,1} Post variable becomes a series of period-level dummies tracking the quarters for the 24 months from April 2015 to March 2017. We include four quarter-level treatment leads for months 1-3 (period -4), 4-6 (period -3), 7-9 (period -2) and 10-12 (period -1) of our study sample period, treatment in months 13-15 (period 0), followed by three treatment lags in 16-18 (period +1) 19-21 (period +2) and 22-24 (period +3) of the sample period. We omit period -1, the quarter before work requirement implementation. The covariates are the same as in the main specification and we cluster the standard errors at the state level.

5. Results:

Table 1 provides descriptive statistics for the outcomes of interest and covariates used in the regression analysis by work requirement waiver status. Although the outcome variables track as expected with the best employment conditions in the states with work requirements throughout the entire time period, our independent variables are fairly consistent across the three types of states. Some obvious differences, e.g., states with work requirements always waived tend to have larger urban populations, support the inclusion of state fixed effects and our control variables.

Figure A1 and A2 display outcome trends by treatment status for the 24 months from April 2015 to March 2017. Figure 1 displays ABAWD labor force participation trends. The trend for the never treated states consistently remains below switcher states and always treated, likely reflecting the relatively poor economic conditions in the never treated states, which qualified for ABAWD waivers for the entire

sample period. The monthly labor force participation rate for never treated states is relatively flat and varies between 60 and 65 percent over the sample period. The trends for switcher states and always treated states overlap and mirror each other, but switcher states display a higher variance and greater vicissitudes. The lowest monthly labor force participation rate for always treated and switcher states are 65 and 60 percent, respectively. The highest point for always treated states is around 75 percent and for switcher states just below 80 percent. Trends for the ABAWD employment variable are given in Figure 2. The never treated trend once again places below switchers and always treated. All three treatment types show similar trends. These graphs support the inclusion of state fixed effects to control for differences in levels across the state groups. Although a pre-existing upward trend in the raw data may exist that reverses shortly after the implementation of work requirements among the switcher states, no obvious discontinuities appear.

Figures A3 and A4 display our outcomes by income category. Labor force participation for the highest income group in our sample ($15,000-$25,000) hovers between 80 to 85 percent, the trend line for those in deep poverty ($0-$7,500) is positioned around the 50 percent mark. Other than the sizable difference in labor force participation for these two low-income groups, the deep poverty sample displays considerably higher variance, implying greater unpredictability or more seasonality in their employment. The trend lines for employment in Figure A4 show a similar disparity by income group, with employment for the $15,000-$25,000 income hovering just below 80 percent, while the trend line for the deep poverty group is around 40 percent and again displaying considerably higher variance. These figures highlight the limited connection the deep poverty group has to the labor market, highlighting concerns that work requirements are unlikely to have a positive employment impact on this group and instead further aggravate their economic plight by phasing them out of SNAP. Figures A5 to A12 depict the outcomes by gender.

Tables 2 and 3 provide regression analysis estimates for single person households making between $0-$7,500 (deep poverty), $7,500-$15,000 (middle poverty), $15,000-$25,000 (around the FPL) and the full sample encompassing $0-$25000 as designated in the column headers. Table 2 analyzes the effect of work requirements on labor force participation and Table 3 evaluates employment effects. Coefficients are reported for the following variables: Post (before or after the second quarter of 2016) and then the interaction of Post with whether a state switched into work requirements (Post * Switchers) and with states that always had work requirements (Post * Always Treated). This second interaction term functions as a naive falsification test, providing the differential impact of “always treated” relative to “never treated” from being in the post period. This interaction coefficient should be statistically indistinguishable from zero, but it is possible that the “always treated” states experience dynamic effects

which may result in failure of the falsification test. Nevertheless, the interaction provides sufficient information to warrant inclusion in the model.

Table 2 shows the results for the overall sample. Only the lowest income sample (earning $7,500 or below) demonstrates a noteworthy statistical relationship; all other estimates are statistically indistinguishable from zero. The overall effect of being in the post-period is a 14.5 percent increase in labor force participation. However, this effect is entirely negated among the states implementing work requirements and partially negated in the always treated states, which may indicate that in addition to an immediate effect, a lagged effect could exist. The set of estimates in Table 3 shows the impact of work requirements on the probability of being employed for the ABAWD sample population. Unlike for labor force participation, the differential impact across state groups appears for the $7,500 to $15,000 income group. Specifically, the $7,500-$15,000 (middle poverty) income sample shows that the implementation of work requirements is associated with a 7.49 percent point increase in the employment outcome ($p<0.01$). The "always treated" interaction term coefficient is statistically indistinguishable from zero, trivial in magnitude and has a negative sign, suggesting lagged effects from work requirements among the always treated group are not contaminating the identified treatment effect.

Table 4 and 5 present the estimates for men. Unlike in Table 2, in Table 4 the overall positive effect on labor force participation is more than negated, rather than merely negated among switcher states, indicating an almost 20 percentage point reduction in labor force participation among the switcher states. Again, a possible lagged effect from the always treated states may exist. Table 5 shows no employment response of work requirement implementation among men.

Tables 6 and 7 posit the estimates for labor force participation and employment outcomes for women. In contrast to our results among male respondents, our results indicate positive effects from work requirements among women. In Table 6, we find that work requirement implementation is associated with a marginally significant 6.42 percent point increase in labor force participation for women in the $7,500-$15,000 (middle poverty) income category, which adds to an overall positive effect on labor force participation within this demographic during the second half of our sample period. The always treated falsification test coefficient has a negative sign and is statistically indistinguishable from zero thus offering robustness to our finding that we are capturing the full effect of the work requirement. In Table 7 we show that work requirement implementation is associated with a 9.10 percentage point increase in the employment for the $7,500-$15,000 income category (Table 7). Again, the "always treated" coefficient is negative and statistically indistinguishable from zero.

The event studies graphs presented in Figures 1-6 evaluate whether the anticipatory or lagged effects are evident. In general, the Figures show limited evidence of consistent negative or positive pre-trends in treatment, although some event studies do exhibit noisy pre-periods, likely due in part to small

sample sizes in some of the underlying regressions. In Figure 1 as in the difference-in-difference results, we see little evidence of significant changes in labor force participation with the implementation of work requirements in the overall sample, but the sub-sample suggest six-month lagged positive effects among the $7,500-$15,000 income sample and negative effects among the $15,000-$25,000 sample.. Given the ABAWD work requirement associated three-month time-limit, lagged effects are expected. For employment in Figure 2, we again find heterogenous trends as a function of income. The $7,500-$15,000 income category shows positive post-treatment coefficients beginning in the period work requirements are implemented, but a lagged negative effect among those earning $15,000-$25,000 beginning about six months post-treatment. Evaluating the event studies by gender in Figures 3-6 indicates that responses to work requirements are driven by women and whether or not the responses are positive varies by income. Women earning between $7,500-$15,000 increase labor force participation and employment following the implementation of work requirements and those earning among $15,000-$25,000 decrease their labor force participation and employment. Men appear to experience no positive effects from work requirements in terms of labor force participation or employment and among the poorest males earning $0 - $7,500, the effect on labor force participation may even be negative.

Conclusion: The stated purpose of associating work requirements with public assistance program is to nudge individuals to rejoin the work force. Using difference-in-differences approaches and an event study, we corroborate the findings in the literature that work requirements have limited impact on overall employment outcomes. We expand the literature by studying labor force participation and how gender affects responses to work requirements. Work requirement imposition increases participation in the labor force and employment conditional on labor force participation for women in the middle-income group ($7,500-$15,000) but has a negative effect on those earning between $15,000 and $25,000 and possibly among the lowest income males as well. The fact that policy responses appear concentrated among women is in line with the literature indicating women are more likely to be involved in public assistance programs (Irving and Loveless, 2015). The varying responses by low-income group may help explain the general limited findings of any employment effect in the literature –women earning between $7,500-$15,000 appear to respond positively to work requirements while those earning between $15,000-$25,000 show evidence of decreased labor force participation and employment in the event studies.

For the lowest income group (income less than $7,500), it appears that the threat and actual loss of food assistance may have no effect on employment outcomes likely because of limited or no links to the labor market. In fact, men in this group may choose to leave the labor market in response to work requirement implementation. The varying responses across the higher low income groups suggest that , the $15,000-$25,000 income group may have already maximized their labor force participation and employment efforts, while those earning between $7,500-$15,000 may have more capacity to change their

labor force participation and employment. Opting out of the labor market is also consistent with opting into or attempting to opt into other forms of public assistance, such as Supplemental Security Income (SSI) or Social Security Disability Insurance (SSDI). Stith (2021) shows work requirements lead to an increase in SSI applications overall and receipt among those with self-reported disabilities. A decrease in labor force participation, with or without a shift towards other forms of public assistance, adds to existing concerns that the most vulnerable individuals may be getting phased out of SNAP and the labor market as a result of ABAWD work requirements.

Implications: The Trump administration pursued a policy of attempting to reduce welfare costs through incentivizing employment, including by reducing the ability of states to obtain SNAP-related ABAWD time limit waivers. In particular, states were limited in their ability to group different regions together in order to apply for a single waiver and the time limit waiver criterion of exceeding 20% of the average U.S. unemployment rate limited to unemployment rates exceeding six percent.[13] The Biden administration, since coming into office, has decided not to implement the aforementioned policy initiatives by the former administration.[14] This highlights the contested and often partisan nature of associating work requirements with public assistance and that the efficacy of attaching work requirements to public assistance recipiency has become a pressing policy question.

Limitations: Given that the implementation of work requirements (treatment) is conditional on labor market conditions and this paper is investigating labor outcomes, the use of a reduced form approach (albeit with identification) raises concerns about endogeneity. It is possible other unincluded labor related factors are correlated with both the outcome of interest and the treatment variable thus biasing our estimates. Mitigating this concern is that work requirement waiver criteria are based on thresholds, measures for which are calculated on a quarterly basis but for a time period encompasses the previous 12 or 24 months depending on the criteria, as opposed to being evaluated on a monthly basis, while we use monthly data for our analysis. This paper also focuses on immediate effects from the 2016 work requirements and may be underestimating effects that persist over time, as suggested by some of the results for states that had work requirements in place at the start of our sample period. Small sample sizes for some of the analyses may limit the robustness of those results, but the general consistency in the effects across specifications supports their inclusion as informative. Our analysis may also be limited because we only consider full implementation of work requirements across an entire state, and thus are unable to evaluate the effects of sub-state partial work requirements. Another potential confounder to our treatment measure includes the ability of states to exempt up to 15 percent of SNAP participants that have

[13] https://www.foodandwine.com/news/snap-food-stamp-work-rules-trump-administration. Accessed 06/17/2021
[14] https://www.foodandwine.com/news/snap-benefits-food-stamps-work-requirement-biden-administration. Accessed 06/17/2021

become ineligible for benefits as a result of the ABAWD time limit. States have discretion in how they implement these exemptions including extending the time-limit for participants who have finished their three-month quota of benefits without meeting the ABAWD work requirement. However, exemptions are unlikely to affect the results as many states with exemptions do not use them, and states that use exemptions typically exempt less than five percent of their ABAWD population (Stith, 2021). Any confounding effect of exemptions would bias our results towards finding no effect. Lastly, this paper does not investigate the first order effect of work requirements on SNAP participation among needy households due to the unavailability of food assistance variables in the CPS – Basic Monthly Survey, and the robustness of reductions in SNAP enrollment found throughout the literature.

To conclude, this paper adds to the literature by showing that work requirements impact labor force participation as well as employment and that these responses vary by gender. In line with the literature (refs – Harris at least), we find these responses vary by income as well. The heterogeneous responses we document provide a clear opportunity for improved policy targeting by income. For example, greater flexibility in meeting work requirements could nudge those in the $7,500-$15,000 bracket to further engage in the work force without discouraging those already more likely to be involved, those in the $15,000-$25,000 income bracket, or those extremely unlikely to be able to become involved, those in the $0-$7,500 bracket. Although in line with prior literature documenting greater accessing of public assistance among women, further research is needed to fully determine the mechanisms behind the gendered responses to work requirements in order to take advantage of the positive effects among women earning between $7,500-$15,000, but avoid the negative effects found among women earning between $15,000-$25,000 and the lowest income males.

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Table 1A: ABAWD Work Requirement Status by State

Work Requirements Never Waived (Never Treated)	*Work Requirements Implemented Q2 2016 (Switchers)*	*Work Requirements Always Waived (Always Treated)*
Delaware	Arkansas	California
Maine	Florida	District of Columbia
Wyoming	Mississippi	Illinois
Iowa	Missouri	Michigan
Nebraska	South Carolina	Nevada
Indiana		New Mexico
Oklahoma		Rhode Island
Kansas		
Texas		

Table 1B: Descriptive Statistics

	Work Requirements Always Waived (Never Treated)	*Two-tailed T-test*	*Work Requirements Waived from Q2 2016 (Switchers)*	*Two-tailed T-test*	*Work Requirements Never Waived (Always Treated)*
Dependent Variables					
Labor Force Participation	0.613 (0.487)	p<0.01	0.631 (0.482)	p<0.01	0.698 (0.458)
Employed	0.553 (0.497)	p<0.01	0.571 (0.494)	p<0.01	0.644 (0.478)
Independent Variables					
White	0.478 (0.500)		0.581 (0.493)		0.676 (0.468)
Hispanic	0.240 (0.427)		0.140 (0.347)		0.137 (0.343)
Black	0.152 (0.359)		0.212 (0.409)		0.108 (0.311)
Native American	0.0198 (0.139)		0.00822 (0.0903)		0.0153 (0.123)

Female	0.456 (0.498)		0.434 (0.496)		0.419 (0.493)
Age	30.47 (8.996)		31.14 (9.138)		29.91 (8.971)
High School	0.273 (0.446)		0.330 (0.470)		0.306 (0.461)
Some College	0.302 (0.459)		0.261 (0.439)		0.314 (0.464)
Associate Degree	0.0780 (0.268)		0.0924 (0.290)		0.0880 (0.283)
Bachelor's Degree	0.178 (0.383)		0.152 (0.359)		0.156 (0.363)
Attending College Part-Time	0.0360 (0.186)		0.0325 (0.177)		0.0352 (0.182)
College Full-Time	0.187 (0.390)		0.144 (0.351)		0.190 (0.393)
Married	0.0310 (0.173)		0.0334 (0.180)		0.0301 (0.171)

Lives in City Center	0.577 (0.494)		0.390 (0.488)		0.261 (0.439)
Number of states-months	216		120		168

Notes: The sample means for individuals in single-person households, between the ages 18-49 and earning $25,000 or less annually.

Table 2: Labor Force Participation Response to ABAWD Work Requirements

Variables	$0-$7,500	$7,500-$15,000	$15,000-$25,000	$0-$25,000
Post	0.145**	-0.00609	0.000124	0.0400
	(0.0614)	(0.0354)	(0.0383)	(0.0292)
Post * Switcher	-0.146*	0.0359	-0.0336	-0.0452
	(0.0705)	(0.0276)	(0.0380)	(0.0292)
Post * Always Treated	-0.126*	-0.0439	0.0299	-0.0344
	(0.0636)	(0.0377)	(0.0357)	(0.0286)
Observations	9,924	9,349	10,057	29,330
R-squared	0.157	0.120	0.108	0.138
Outcome Mean	0.502	0.650	0.817	0.659

Notes: Linear probability model regression with data from the Current Population Survey. All regressions include state and month fixed effects and demographic control variables including race/ethnicity, educational attainment, marital status, and urban/rural. Standard errors corrected for clustering on the state are in parentheses. The sample consists of ages 18-49, without dependents.
*** Significant at 1% level ** at 5% level * at 10% level

Table 3: Employment Response to ABAWD Work Requirements

Variables	$0-$7,500	$7,500-$15,000	$15,000-$25,000	$0-$25,000
Post	0.0954	-0.0429	-0.00294	0.0151
	(0.0615)	(0.0304)	(0.0306)	(0.0320)
Post * Switcher	-0.108	0.0749***	-0.0292	-0.0257
	(0.0675)	(0.0210)	(0.0240)	(0.0284)
Post * Always Treated	-0.0903	-0.0132	0.0299	-0.0172
	(0.0602)	(0.0347)	(0.0268)	(0.0268)
Observations	9,768	9,207	9,875	28,850
R-squared	0.118	0.106	0.090	0.120
Outcome Mean	0.413	0.602	0.787	0.602

Notes: Linear probability model regression with data from the Current Population Survey. All regressions include state and month fixed effects and demographic control variables including race/ethnicity, educational attainment, marital status, and urban/rural. Standard errors corrected for clustering on the state are in parentheses. The sample space consists of ages 18-49, without dependents, earning less than $25,000 per year.
*** Significant at 1% level ** at 5% level * at 10% level

Table 4: Labor Force Participation Response to ABAWD Work Requirements - Men

Variables	$0-$7,500	$7,500-$15,000	$15,000-$25,000	$0-$25,000
Post	0.164**	-0.0466	0.0288	0.0442
	(0.0669)	(0.0554)	(0.0425)	(0.0278)
Post * Switcher	-0.199**	-0.00259	-0.0209	-0.0737**
	(0.0901)	(0.0494)	(0.0398)	(0.0329)
Post * Always Treated	-0.160*	-0.0561	0.0373	-0.0453
	(0.0813)	(0.0535)	(0.0387)	(0.0303)
Observations	5,191	5,221	5,821	16,233
R-squared	0.207	0.168	0.137	0.183
Outcome Mean	0.490	0.666	0.837	0.671

Notes: Linear probability model regression with data from the Current Population Survey. All regressions include state and month fixed effects and demographic control variables including race/ethnicity, educational attainment, marital status, and urban/rural. Standard errors corrected for clustering on the state are in parentheses. The sample space consists of ages 18-49, without dependents, earning less than $25,000 per year.
*** Significant at 1% level ** at 5% level * at 10% level

Table 5: Employment Response to ABAWD Work Requirements - Men

Variables	$0-$7,500	$7,500-$15,000	$15,000-$25,000	$0-$25,000
Post	0.0877	-0.0793	0.0201	0.0123
	(0.0899)	(0.0574)	(0.0477)	(0.0421)
Post * Switcher	-0.127	0.0307	-0.0254	-0.0471
	(0.0956)	(0.0501)	(0.0349)	(0.0395)
Post * Always Treated	-0.106	-0.0444	0.0306	-0.0324
	(0.0820)	(0.0539)	(0.0326)	(0.0315)
Observations	5,122	5,148	5,742	16,012
R-squared	0.164	0.147	0.109	0.157
Outcome Mean	0.398	0.606	0.800	0.609

Notes: Linear probability model regression with data from the Current Population Survey. All regressions include state and month fixed effects and demographic control variables including race/ethnicity, educational attainment, marital status, and urban/rural. Standard errors corrected for clustering on the state are in parentheses. The sample space consists of ages 18-49, without dependents, earning less than $25,000 per year.
*** Significant at 1% level ** at 5% level * at 10% level

Table 6: Labor Force Participation Response to ABAWD Work Requirements - Women

Variables	$0-$7,500	$7,500-$15,000	$15,000-$25,000	$0-$25,000
Post	0.129*	0.0858*	-0.0358	0.0376
	(0.0632)	(0.0438)	(0.0553)	(0.0387)
Post * Switcher	-0.0732	0.0642*	-0.0505	-0.00916
	(0.0796)	(0.0358)	(0.0470)	(0.0362)
Post * Always Treated	-0.0869	-0.0693	0.0245	-0.0183
	(0.0587)	(0.0450)	(0.0404)	(0.0250)
Observations	4,733	4,128	4,236	13,097
R-squared	0.144	0.125	0.099	0.108
Outcome Mean	0.515	0.644	0.790	0.645

Notes: Linear probability model regression with data from the Current Population Survey. All regressions include state and month fixed effects and demographic control variables including race/ethnicity, educational attainment, marital status, and urban/rural. Standard errors corrected for clustering on the state are in parentheses. The sample space consists of ages 18-49, without dependents, earning less than $25,000 per year.

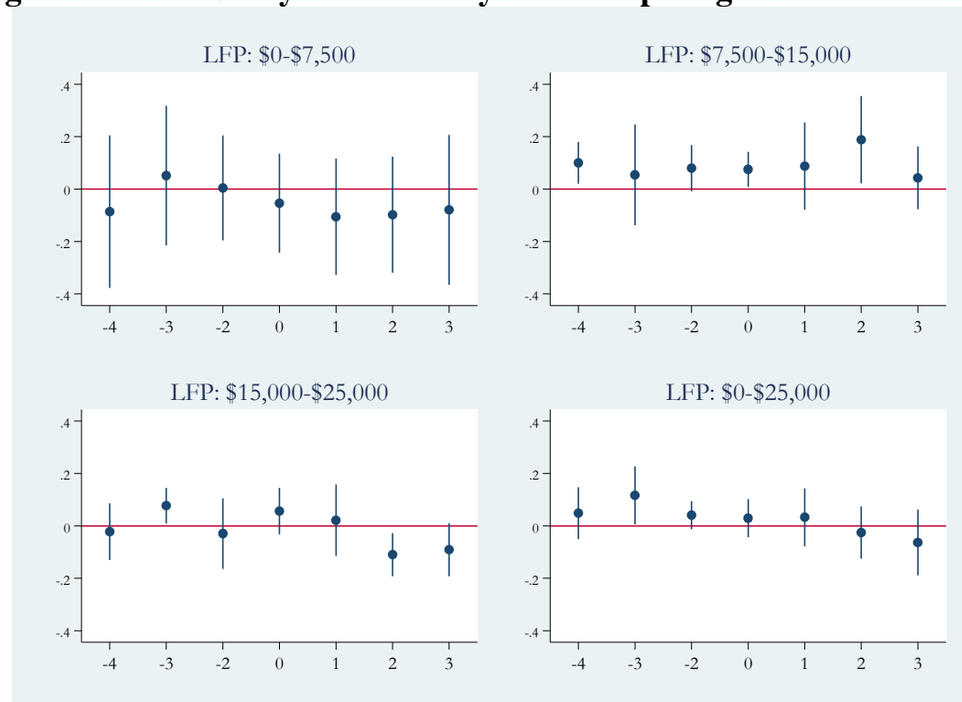
*** Significant at 1% level ** at 5% level * at 10% level

Table 7: Employment Response to ABAWD Work Requirements - Women

Variables	$0-$7,500	$7,500-$15,000	$15,000-$25,000	$0-$25,000
Post	0.103*	0.0522	-0.0363	0.0182
	(0.0590)	(0.0432)	(0.0585)	(0.0359)
Post * Switcher	-0.0750	0.0910**	-0.0260	0.00168
	(0.0748)	(0.0345)	(0.0380)	(0.0320)
Post * Always Treated	-0.0722	-0.0268	0.0367	0.00698
	(0.0522)	(0.0446)	(0.0331)	(0.0233)
Observations	4,646	4,059	4,133	12,838
R-squared	0.115	0.120	0.091	0.099
Outcome Mean	0.431	0.598	0.771	0.593

Notes: Linear probability model regression with data from the Current Population Survey. All regressions include state and month fixed effects and demographic control variables including race/ethnicity, educational attainment, marital status, and urban/rural. Standard errors corrected for clustering on the state are in parentheses. The sample space consists of ages 18-49, without dependents, earning less than $25,000 per year.
*** Significant at 1% level ** at 5% level * at 10% level

Figure 1: Event Study – Probability of Participating in the Labor Force

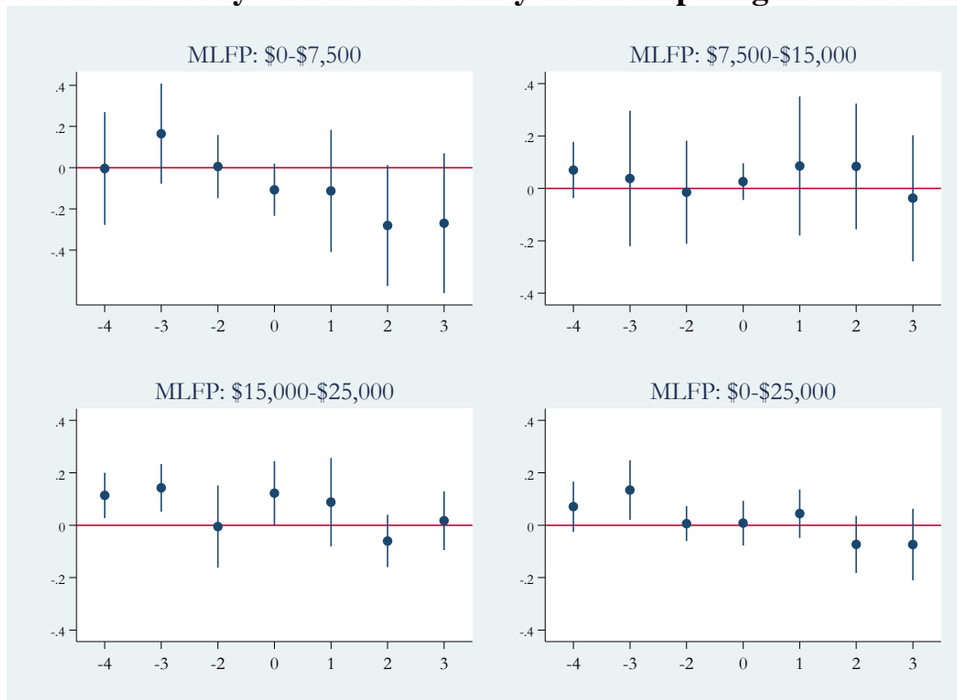


Notes: The sample consists of single person households, ages 18-49, without dependents. Respondents are in income category $0-$25000. The x-axis shows months relative to work requirements, implemented at time t=0 (the second quarter of 2016) and the y-axis shows average treatment effects among the treated.

Figure 2: Event Study – Employment

Notes: The sample consists of single person households, ages 18-49, without dependents. Respondents are in income category $0-$25000. The x-axis shows months relative to work requirements, implemented at time t=0 (the second quarter of 2016) and the y-axis shows average treatment effects among the treated.

Figure 3: Event Study – Male Probability of Participating in the Labor Force

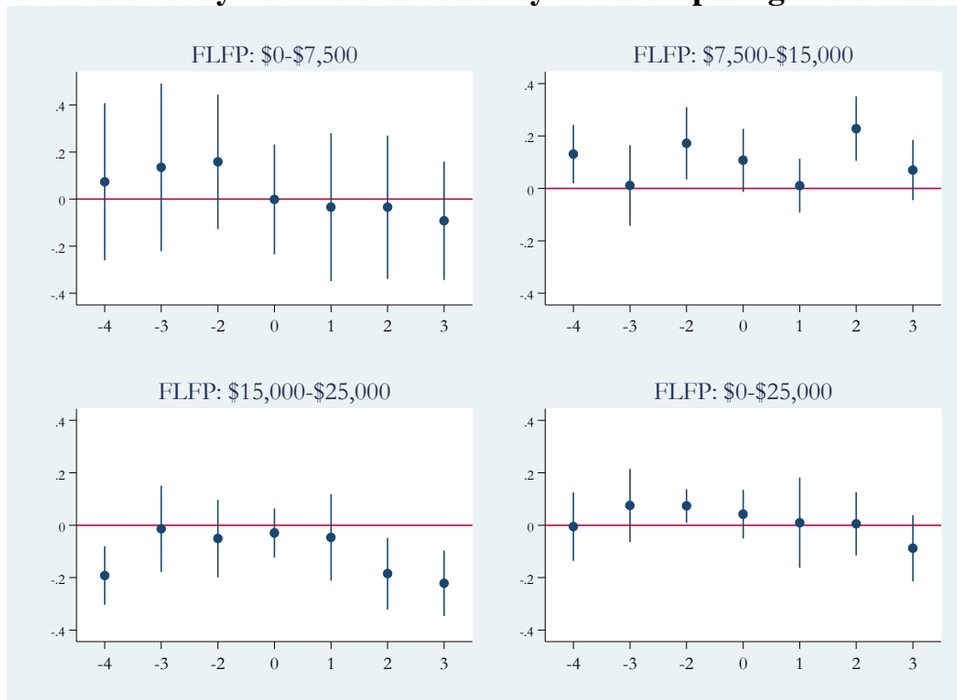


Notes: The sample consists of single person households, ages 18-49, without dependents. Respondents are in income category $0-$25000. The x-axis shows months relative to work requirements, implemented at time t=0 (the second quarter of 2016) and the y-axis shows average treatment effects among the treat

Figure 4: Event Study – Male Employment

Notes: The sample consists of single person households, ages 18-49, without dependents. Respondents are in income category $0-$25000. The x-axis shows months relative to work requirements, implemented at time t=0 (the second quarter of 2016) and the y-axis shows average treatment effects among the treated.

Figure 5: Event Study – Female Probability of Participating in the Labor Force



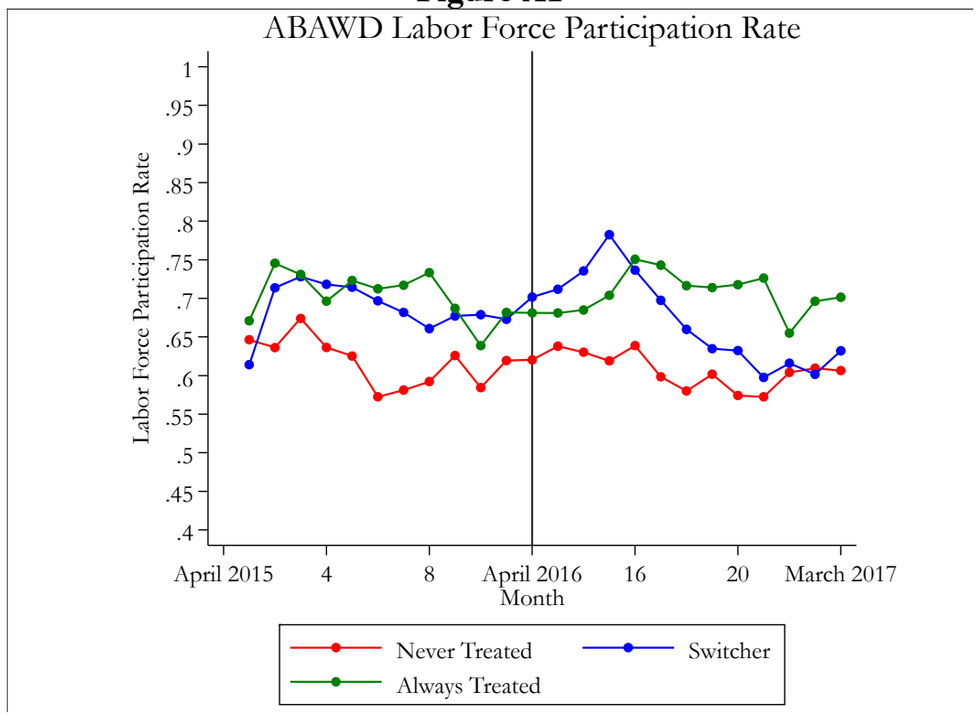
Notes: The sample consists of single person households, ages 18-49, without dependents. Respondents are in income category $0-$25000. The x-axis shows months relative to work requirements, implemented at time t=0 (the second quarter of 2016) and the y-axis shows average treatment effects among the treated.

Figure 6: Event Study – Female Employment

Notes: The sample consists of single person households, ages 18-49, without dependents. Respondents are in income category $0-$25000. The x-axis shows months relative to work requirements, implemented at time t=0 (the second quarter of 2016) and the y-axis shows average treatment effects among the treated

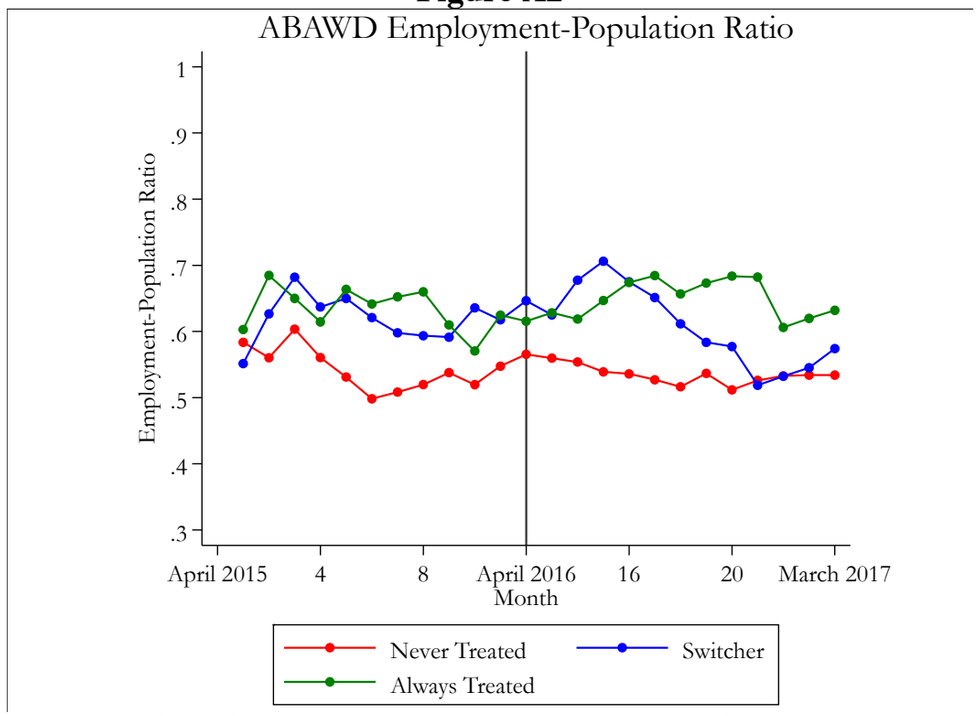
Appendix A

Figure A1



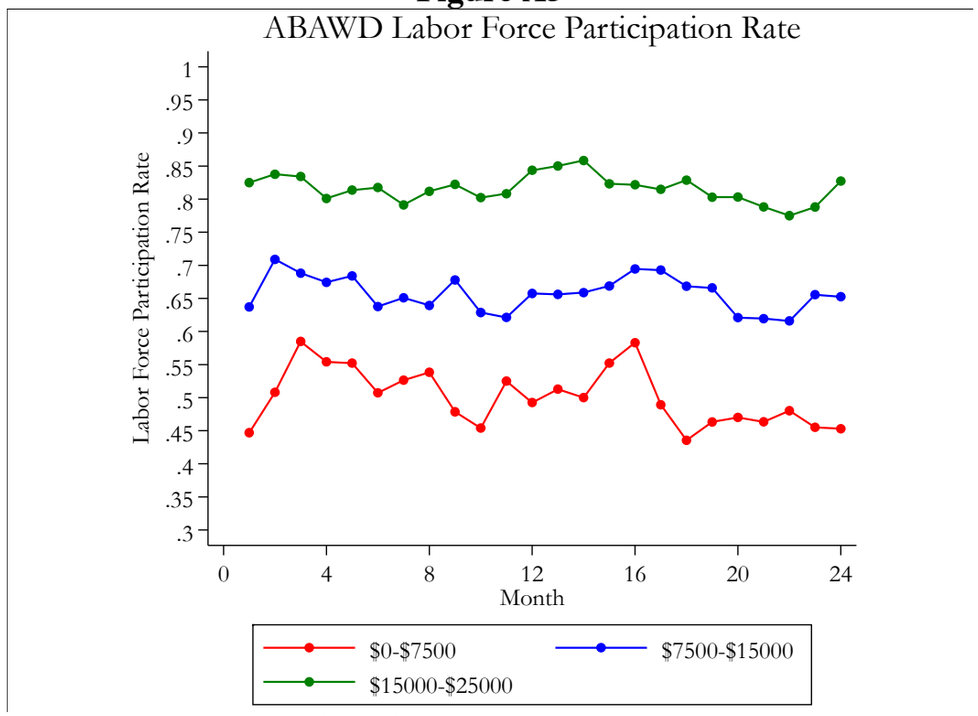
Notes: The sample consists of single person households, ages 18-49, without dependents. Respondents are in income category $0-$25000.

Figure A2



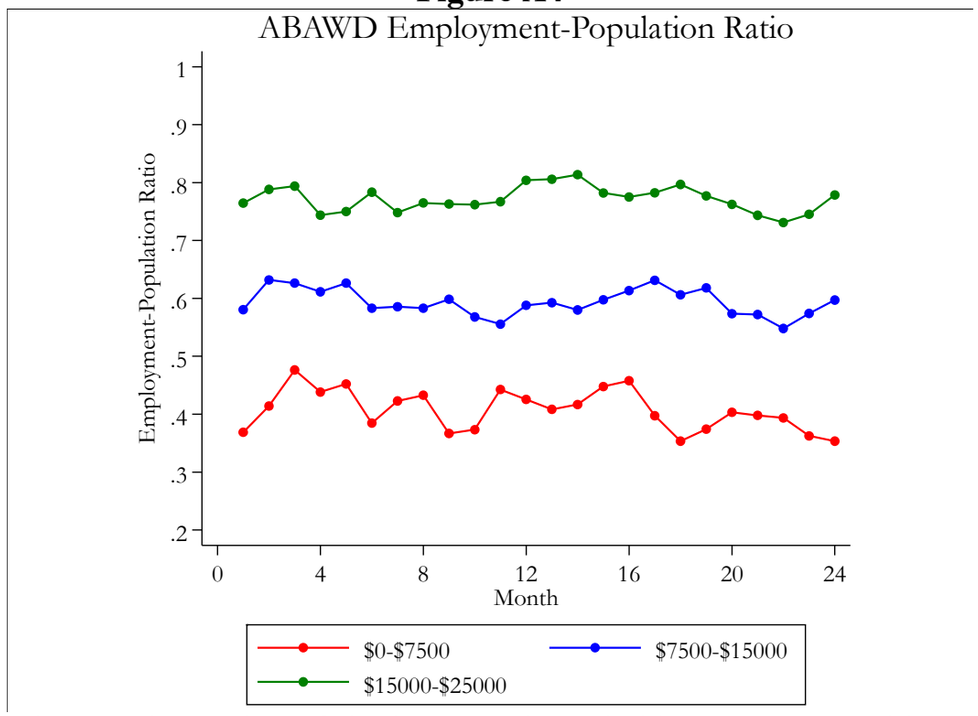
Notes: The sample consists of single person households, ages 18-49, without dependents. Respondents are in income category $0-$25000.

Figure A3



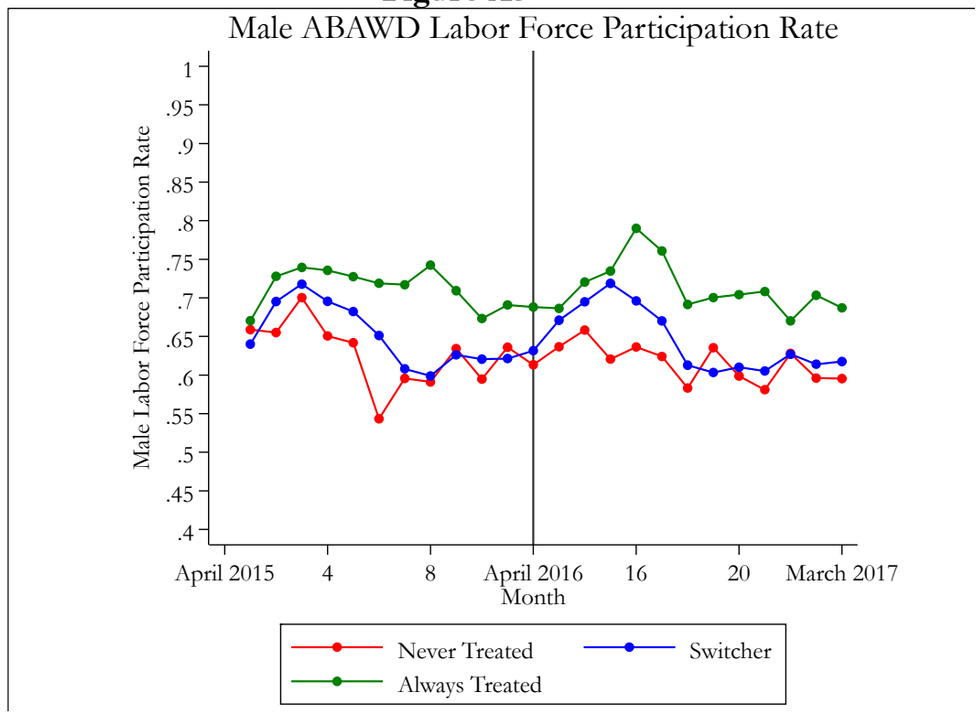
Notes: The sample consists of single person households, ages 18-49, without dependents.

Figure A4



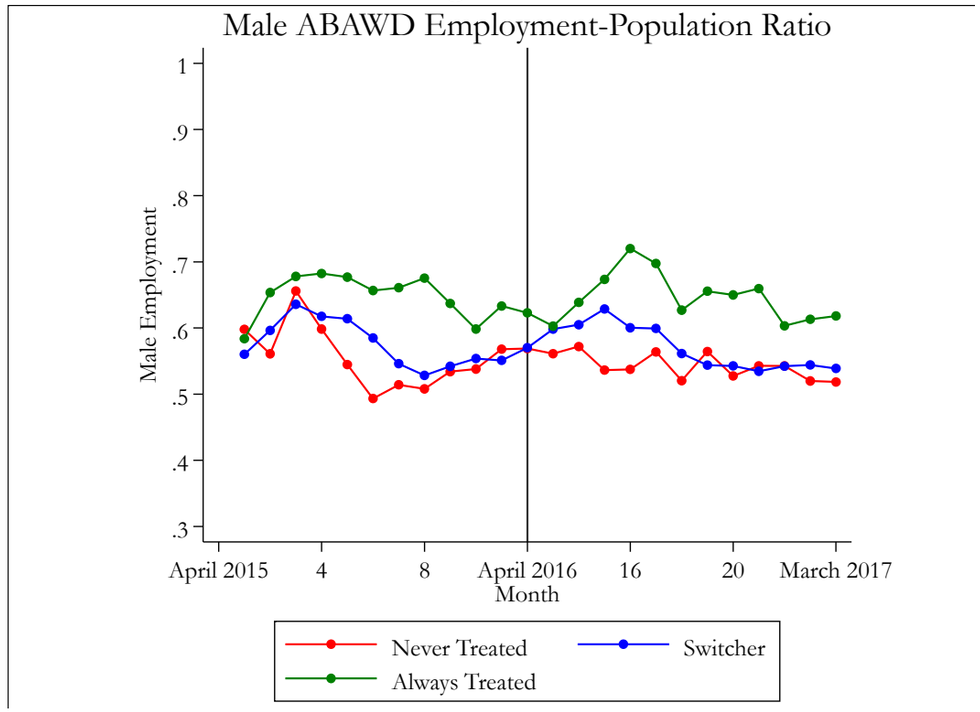
Notes: The sample consists of single person households, ages 18-49, without dependents.

Figure A5



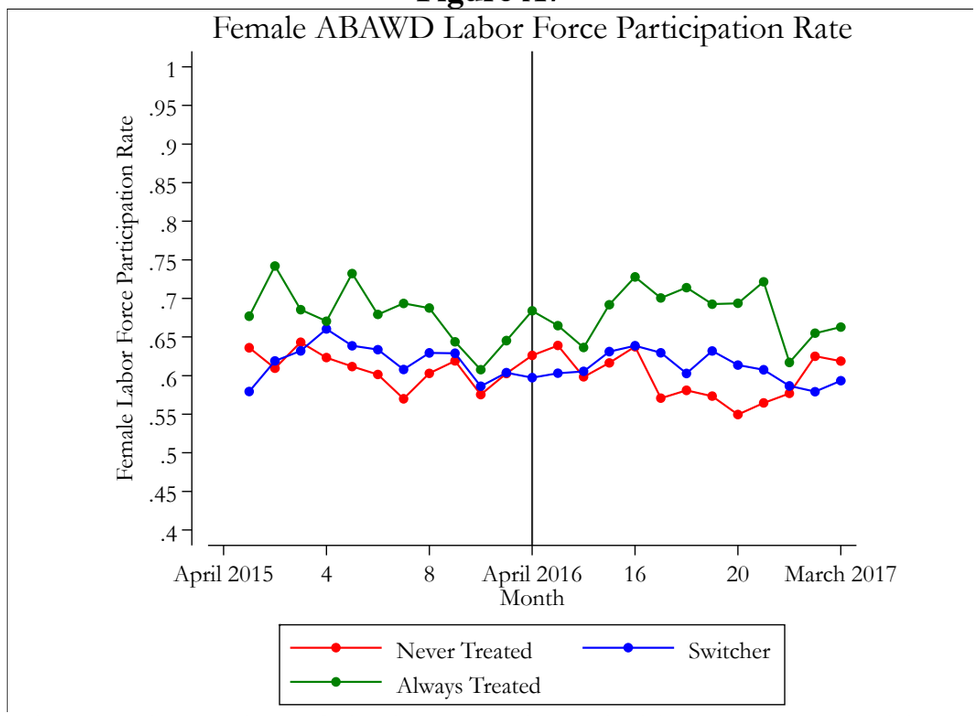
Notes: The sample consists of single person households, ages 18-49, without dependents.

Figure A6



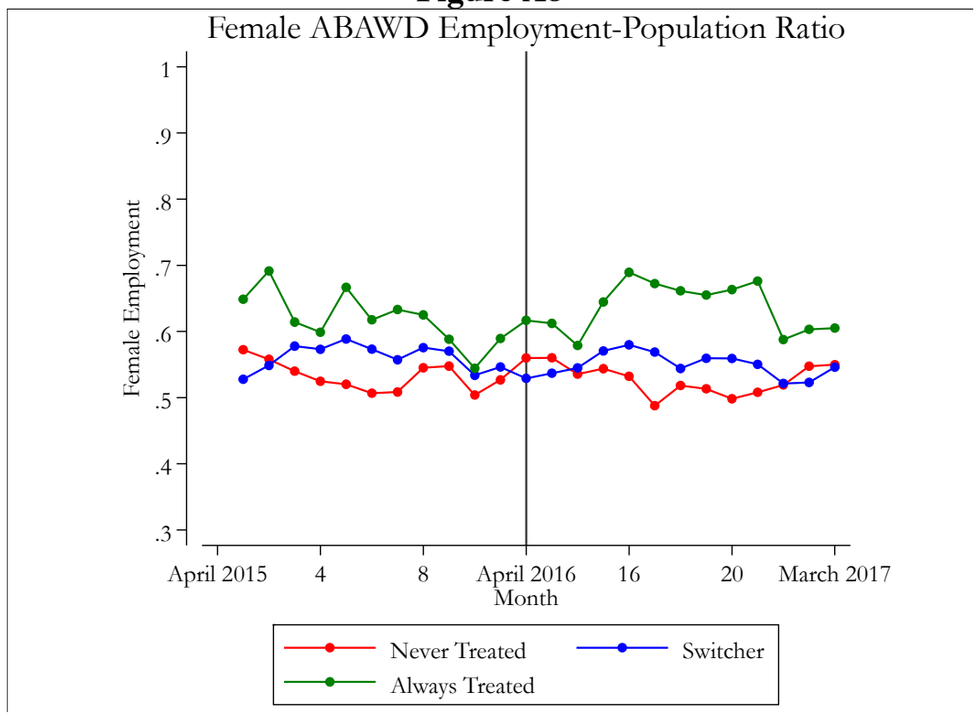
Notes: The sample consists of single person households, ages 18-49, without dependents.

Figure A7



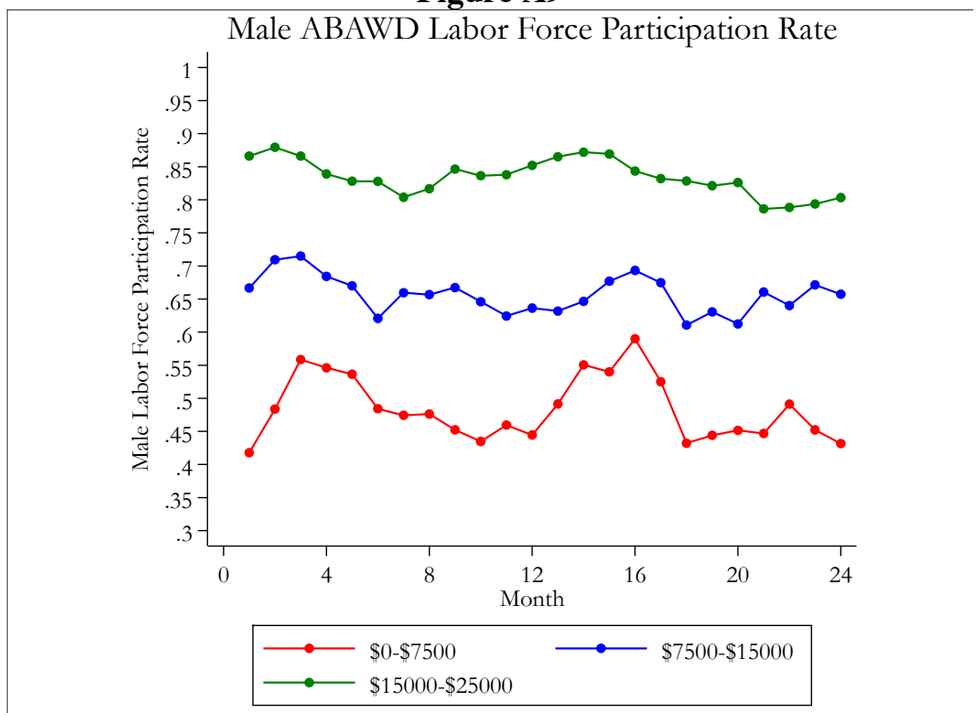
Notes: The sample consists of single person households, ages 18-49, without dependents.

Figure A8



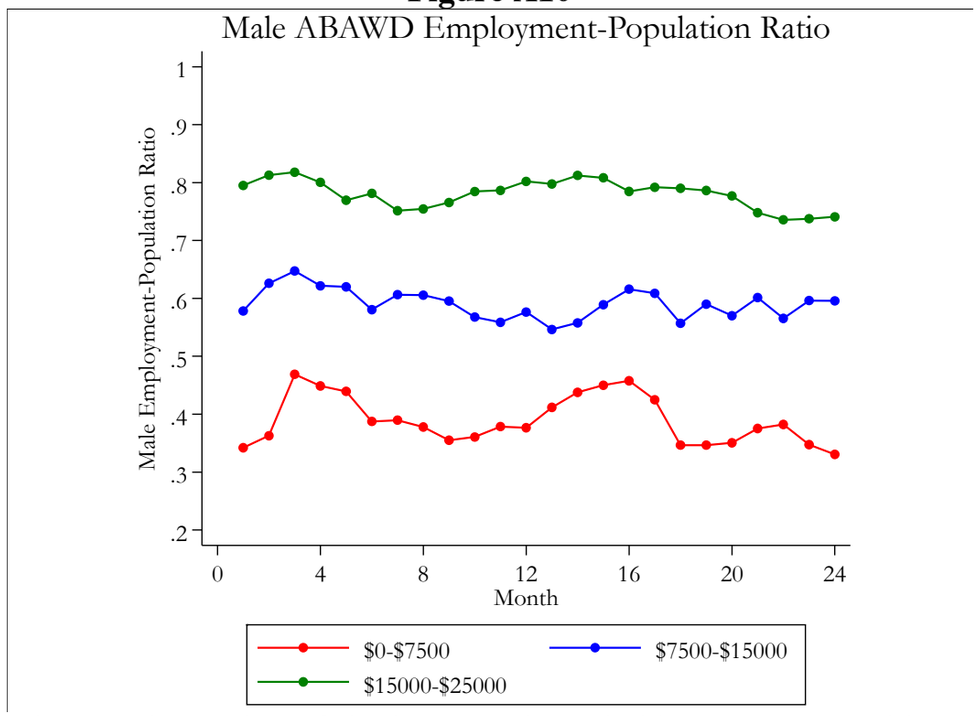
Notes: The sample consists of single person households, ages 18-49, without dependents.

Figure A9



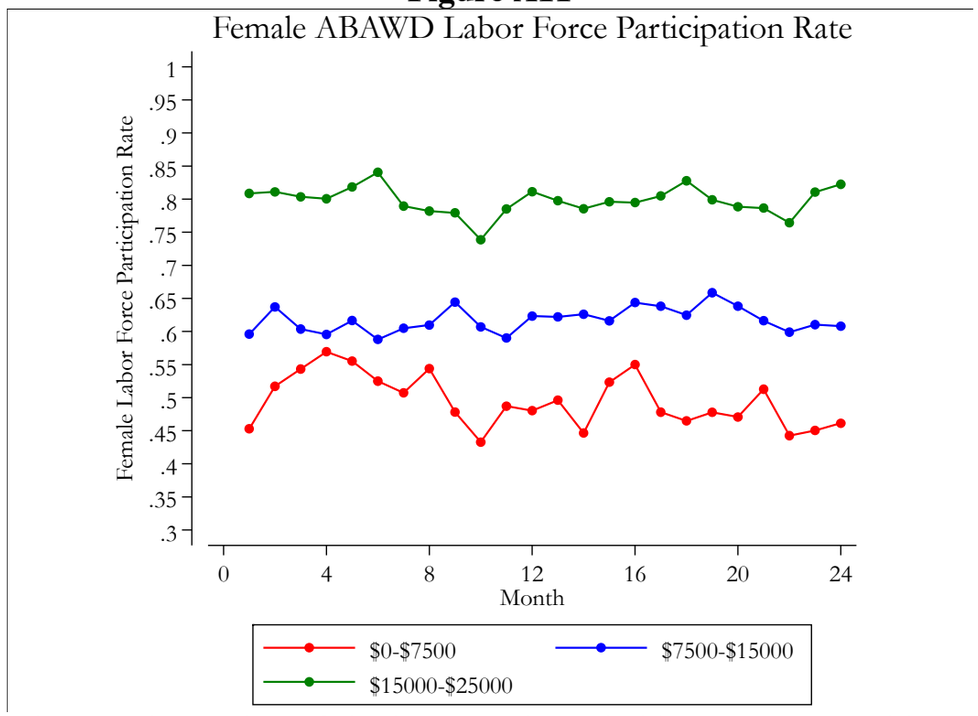
Notes: The sample consists of single person households, ages 18-49, without dependents.

Figure A10



Notes: The sample consists of single person households, ages 18-49, without dependents.

Figure A11



Notes: The sample consists of single person households, ages 18-49, without dependents.

Figure A12

Notes: The sample consists of single person households, ages 18-49, without dependents.